Task-1: Data Analysis.
Analysis Report.

Note: Using this report along with dashboard is recommended.

1. Dataset Overview

Object classes: 10

Training Images	Validation Images	Total Images
69863	10000	79863

Training Annotations	Validation Annotations	Total Annotations
1286871	185526	1472397

Average annotation per image in training	Average annotation per image in validation	Average annotation per image in overall dataset.
18.41	18.55	18.43

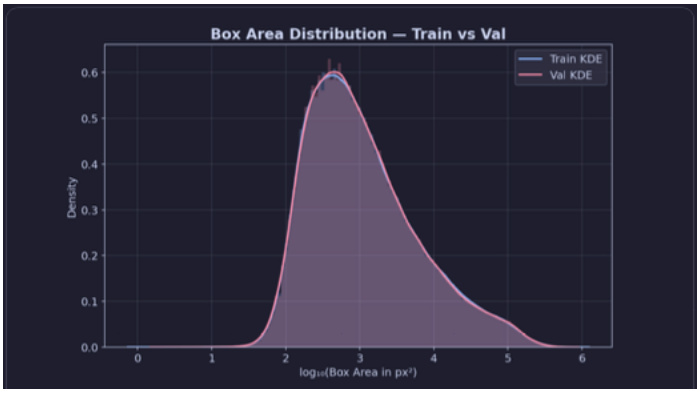
Inferences:

i. Dataset follows train and validation split of 87.47% and 12.52%.
ii. On an average 18 annotations per image are present suggesting dense object detection environments.
iii. The quantity of validation data is good to evaluate the model after training with good quantity of data.

2. Train Validation split and Distribution analysis (Overall).

A. Please refer following plot from the dashboard:

[Split Analysis & Distributions -> Box Area Distribution — Train vs Val]

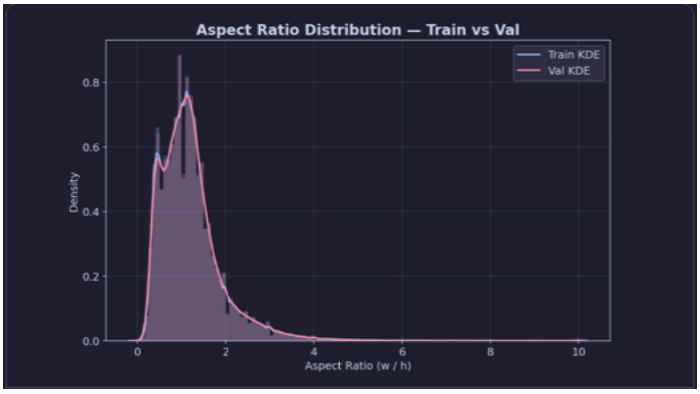


Inferences:

i. Curves overlap perfectly indicating no size distribution shift in training and validation.
ii. Distribution peaks at around 2.5 to 3 values of box area (x-axis) indicating it's a dominant size in both the dataset.
iii. Presence of a tail with areas smaller than 2 indicate presence of very small object which may act as noise during detection. (Filtering then out will help to train better)
iv.

B. Please refer following plot from the dashboard:

[Split Analysis & Distributions -> Aspect Ratio Distribution — Train vs Val]



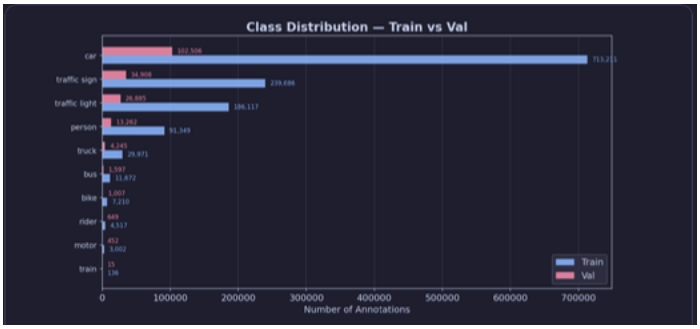
Inferences:

i. Curves overlap perfectly indicating geometric similarity between training and validation data.

ii. Peak shows most of the objects are around the value of 1 indicating most of the objects are of nearly square shape (close up cars, traffic lights etc). less than 1 showing long objects like pedestrians or bikes etc lastly after 2 large broad (width more) object are present:
Presence: Near square objects > Long objects (more height) > Broad objects (more width).

C. Please refer following plot from the dashboard:

[Split Analysis & Distributions -> Class Distribution — Train vs Val]



Inferences:

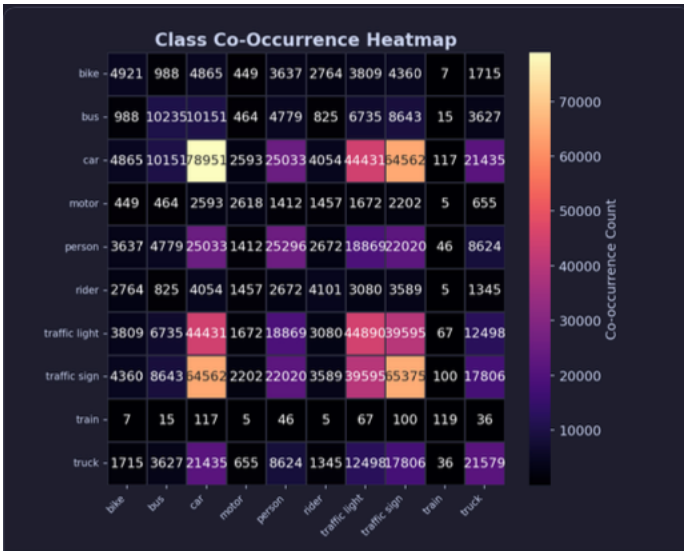
i. Class imbalance Is present in which Car dominates the object presence by a huge margin.
"Car > traffic sign > traffic light > person > truck > bus > Bike > rider > motor>train."
Same distribution is present for both training and validation dataset.

ii. Even for the rare class split is consistent.

iii. Using dataset In its current form model may not perform well with classes having less sample.

D. Please refer following plot from the dashboard:

[Split Analysis & Distributions -> Class Co-Occurrence Heatmap]



Inferences:

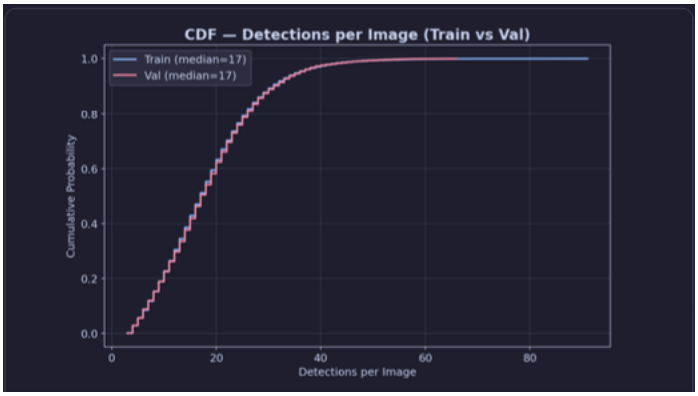
i. High co-occurrence of car with other cars traffic light, traffic signs etc. Which is expected. (Car, traffic light and signs are having highest samples).

ii. Rider class appears most of the time with other vehicle related classes indicating a co-relation.

iii. Even the rare classes are present mixing with other classes indicating good mix of complex scenario.

E. Please refer following plot from the dashboard:

[Split Analysis & Distributions -> CDF — Detections per Image]



Inferences:

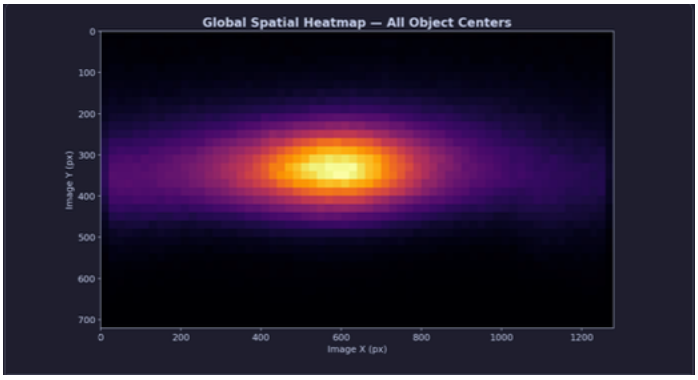
i. Overlapping of train and validation plot shows statistical similarity between the scenes. Training performance will be accurately reflected by validation metrics.

ii. Both shows a median of around 17 detections per image. (close to average of 18).

iii. Training data seem to contain some more complex scenes (tail over ~65 detections per image).

F. Please refer following plot from the dashboard:

[Split Analysis & Distributions -> Global Spatial Heatmap]



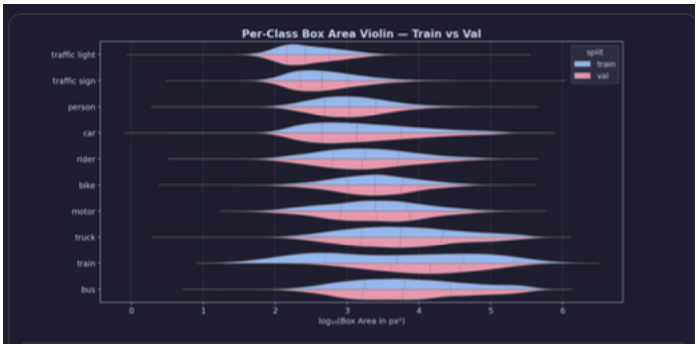
Inferences:

i. Strong bright point at the center represents Vanishing point of mounted camera. Most of the Class objects will converse in this area which is natural.

ii. The black areas on top (because of sky) and bottom (because of vehicles hood etc) will bring bias to the trained vehicle. Indicating dead zones. (Augmentation needed)

G. Please refer following plot from the dashboard:

[Split Analysis & Distributions -> Per-Class Box Area Violin — Train vs Val]



Inferences:

i. Train and validation size distributions mirror perfectly across all classes

ii. Object sizes range drastically both between classes (tiny lights vs. huge buses) and within them (distant cars vs. close cars).

iii. Extreme size outliers are proportionally equal across both splits, though the larger training set holds a higher absolute count of these edge cases.

3. Train Validation split and Distribution analysis (Class-wise).

Class	Train Annotation.	Val Annotation.	Total	Aspect Ratio	Box Size	Spatial Pattern	% of whole data annotation
Car	7,13,211	1,02,506	8,15,717	~1.8	Medium–Large	Horizon band	55
Traffic Sign	2,39,686	34,908	2,74,594	~1.0	Small–Medium	Upper-center	19
Traffic Light	1,86,117	26,885	2,13,002	~0.5	Small	Upper-center	14
Person	91,349	13,262	1,04,611	~0.5	Small–Medium	Lower-center	7
Truck	29,971	4,245	34,216	~1.5	Large	Horizon band	2
Bus	11,672	1,597	13,269	~1.3	Large	Center	1

Rider	4,517	649	5,166	~0.75	Small–Medium	Center	0
Bike	7,210	1,007	8,217	~0.9	Small	Scattered	1
Motor	3,002	452	3,454	~1.0	Small	Scattered	0
Train	136	15	151	~2.5+	Variable	Sparse	0.01025539
Total	12,86,871	1,85,526	14,72,397				100

Class	Key Observations
Car	• Present as square /wide box. • More densely packed in single frame (multiple detections/frequent). • Detections are concentrated in a central horizontal vanishing point band (far objects look centrally situated later spreading out).
Traffic Sign	• Predominantly present as very small, square objects (with the aspect ratio peaking sharply at 1.0). • The spatial heatmap shows a massive concentration in a wide, upper-central band across the image, logically corresponding to where road signs are mounted overhead.
Traffic Light	• Traffic lights consistently appear as exceptionally small, tall-and-narrow objects (with the aspect ratio sharply peaking below 0.5). • The spatial heatmap reveals a highly intense, singular concentration in the upper-center of the frame.
Person	• People consistently appear as highly vertical, tall-and-narrow objects (with the aspect ratio sharply peaking around 0.4). • The spatial heatmap for people reveals a distinct bi-modal twin peak pattern, confirming they predominantly appear on the left and right sides of the frame (sidewalks/crosswalks) rather than the direct center of the road.
Truck	• Trucks predominantly present as slightly wide or square objects (with the aspect ratio peaking just above 1.0) and typically appear sparsely as 1 to 3 instances per image. • Detections are densely clustered along the central horizontal vanishing-point band, with larger bounding boxes reliably appearing lower in the frame as they approach the camera.
Rider	• Riders consistently manifest as distinctly tall-and-narrow objects (with the aspect ratio peaking sharply around 0.5) and typically appear sparsely as just 1 or 2 instances per frame. • Detections are densely clustered along a central horizontal band across the image.
Bike	• Bikes predominantly manifest as slightly tall-and-narrow objects (aspect ratio < 1), though variations like side-profiles create a distinct tail of wider bounding boxes. • Bikes generally appear sparsely (1-2 instances per image) . • Spatially concentrated in a central horizontal band (more spread evenly in this), avoiding the top and bottom edges of the dashcam frame.
Motor	• Motorcycles typically present as slightly tall-and-narrow objects (aspect ratio peaking just below 1.0) and almost always appear as solitary instances within a frame. • Broadly clustered across the middle horizontal band. • Larger boxes appears as they approach ego vehicle.

Train	• Due to the extreme lack of data (only 151 total annotations), the training and validation distributions are jagged and misaligned, warning of severe evaluation volatility and domain shift. • Trains exhibit massive variance in area and aspect ratio (featuring heavily wide, multimodal peaks) and almost exclusively appear as solitary, massive instances within a frame. • Lacking the standard central vanishing-point concentration.
Bus	• Predominantly present as tall-and-narrow objects (aspect ratio < 1), with wider bounding boxes occasionally capturing side-profile views. • Heavily concentrated in a central horizontal band (more on mid)

4. Per-Class Anomalies & Edge Cases.

Anomalies and edge cases are selected on the basis of 'Largest', 'Smallest', 'Tallest', 'Widest', 'Position', 'Outlier' and 'Crowded'.

To visualise and know more details please visit the Dashboard

Class	Anomalies & Edge Cases
Car	17
Traffic (Combined)	31
Person	19
Truck	20
Rider	12
Bike	17
Motor	20
Train	19
Bus	21